# Jacopo Di Silvestre

Health Economic Modeling | HEOR | Cost-Effectiveness Analysis

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## Professional Summary

Health economist and PhD researcher specializing in decision-analytic modeling, microsimulation, and cost-effectiveness analysis in oncology. Experienced in developing and calibrating C++ disease simulation models and conducting economic evaluations and statistical analyses in R. Research collaborations with Mass General Brigham Center for Health Technology Assessment, with peer-reviewed publications in *The Lancet Regional Health – Europe* and *Annals of Internal Medicine*.

## Technical Skills

**Health Economic Modeling:** Cost-effectiveness and cost-utility analysis, individual-level microsimulation, decision-analytic modeling, model calibration and validation, sensitivity analyses

**Quantitative Methods:** Statistical modeling, regression analysis, epidemiological modeling, survival analysis, evidence synthesis

**Programming:** C++, R, Excel, Bash, Git, Visual Studio, Slurm, High-Performance Computing (HPC)

## Research Experience

**PhD Candidate** Mar 2025 - Present

University of Oslo, Oslo, Norway

- Develop and calibrate individual-level lung cancer microsimulation models in C++ to evaluate alternative screening strategies, comparing eligibility criteria, screening intervals, clinical outcomes, and cost-effectiveness to inform Norwegian screening policy.
- Estimate age-, sex-, and birth cohort-specific smoking behaviors using Norwegian population survey data to inform individual-level simulations.
- Conduct model-based economic evaluations, including uncertainty and scenario analyses, and investigate the integration of biomarkers, AI-based risk prediction, and smoking cessation interventions into screening strategies.

**Research Scholar** Nov 2024 - Present

Harvard Medical School - Massachusetts General Hospital
Center for Health Technology Assessment, Boston, MA

- Development and calibration of the Lung Cancer Policy Model (LCPM), a CISNET microsimulation model, to support analyses in the United States and Norway.
- Testing C++ simulation code, assess model performance, and analyze simulation outputs.

**Research Assistant** Jul 2024 - Feb 2025

University of Oslo, Oslo, Norway

- Developed an individual-based lung cancer natural history model in R to evaluate the cost-effectiveness of risk-based low-dose CT screening in a high-risk Norwegian population.
- Analyzed outputs from the CISNET Harvard-CC model in R for a cost-effectiveness study of cervical cancer screening strategies in Norway.
- Conducted a systematic review of the health impact and cost-effectiveness of one-dose human papillomavirus (HPV) vaccination compared with alternative schedules (Mark Jit, Jane J Kim, Jean-François Laprise, Marc Brisson, Kiesha Prem, Emily A Burger).

## Research Experience (continued)

**Research Intern** Sep 2023 - Feb 2024
ASCERTAIN European Project

- Conducted a scoping review of statistical methods for estimating comparative clinical effectiveness using evidence from single-arm or non-randomized trials (Eline Aas, Emily A Burger).

## Selected Publications

**Di Silvestre J**, Salgado MV, Jeon J, Mok Y, Holford TR, Vedøy TF, Pedersen K, Burger EA, Meza R. Age- and sex-specific smoking trends by birth cohort in Norway, 1978-2024: an analysis of cross-sectional survey data. *The Lancet Regional Health - Europe*. 2026;71:101886.
doi: 10.1016/j.lanepe.2026.101886

Pedersen K, **Di Silvestre J**, Sy S, Portnoy A, Castle PE, Kim JJ, Burger EA. Optimizing Cervical Cancer Screening by Age at Vaccination for Human Papillomavirus: Health and Resource Implications. *Annals of Internal Medicine*. 2026.
doi: 10.7326/ANNALS-25-03192

## Education

**PhD Studies** Expected 2028
University of Oslo, Norway

**MS in Health Economics and Health Management** Sep 2022 - Jun 2024
Erasmus University Rotterdam

- European Master in Health Economics and Management (EU-HEM), with exchange studies at the University of Oslo and the University of Bologna.
- **Thesis:** "Risk-Based Lung Cancer Screening: A Microsimulation Model for the Norwegian High-Risk Population.", supervisors Emily A. Burger and Natalia Kunst
- Through an individual-based natural history model of lung cancer, this thesis analysis estimated NOK 129,689 per quality-adjusted life-year (QALY) gained for the evaluated risk-based screening strategy versus no screening in the modeled high-risk Norwegian population.
- **Relevant coursework:** Advanced Health Economic Modeling; Health Technology Assessment; Pharmaceutical Pricing and Market Access; Econometrics; Discrete Choice Experiments.

**BS in Economics** 2019 - 2022
University of Pavia, Italy
Grade: 109/110.

## Teaching and Supervision

Mentor master's students conducting health economic modeling projects in R and Excel, providing guidance on model development, literature synthesis, clinical treatment pathways, and cost estimation. Oversee methodological decisions and review technical outputs throughout thesis development.